

company sold transgenic plants resistant to the company's herbicides, but that could not produce their own seed, forcing the farmer to buy new seed each year. An international outcry led to the abandonment of this practice, but the use of crops that are resistant to certain herbicides, with a single company owning both seed and herbicide, continues. Some see this as a conflict of interest.

Groups that oppose genetically modified foods sometimes behave unethically. In 1999 environmental activists destroyed an experimental forest of poplars near London. The trees were indeed transgenic, but the experiments were designed to see if the trees would require fewer chemical herbicides, an activity the environmentalists had themselves suggested. More alarming were several incidents in the United States in 2000, when people who object to genetically modified foods vandalized laboratories and destroyed fields of crops, some of which were not even transgenic.

So far, foods containing GMOs appear to be safe. They may be easier to cultivate and may permit the development of new variants. However, it will take more time to determine whether or not they have longer term health and ecological effects. SEE ALSO BIOTECHNOLOGY: ETHICAL ISSUES; GENETIC TESTING: ETHICAL ISSUES; GENETICALLY MODIFIED FOODS; TRANSGENIC ANIMALS; TRANSGENIC PLANTS.

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Transgenic Plants

Transgenic plants are plants that have been genetically modified by inserting genes directly into a single plant cell. Transgenic crop plants modified for improved flavor, pest resistance, or some other useful property are being used increasingly.

Transgenic plants are unique in that they develop from only one plant cell. In normal sexual reproduction, plant offspring are created when a pollen cell and an ovule fuse. In a similar laboratory procedure, two plant cells that have had their cell walls removed can be fused to create an offspring.

Genetic Engineering Techniques

There are three general approaches that can be used to insert the DNA into a plant cell: **vector**-mediated transformation, particle-mediated transformation, and direct DNA absorption. With vector-mediated transformation, a plant cell is infected with a virus or bacterium that, as part of the infection process, inserts the DNA. The most commonly used vector is the crown-gall bacterium, *Agrobacterium tumefaciens*. With particle-mediated transformation (particle bombardment), using a tool referred to as a "gene gun," the DNA is carried into the cell by metal particles that have been acceler-

vector carrier



ated, or “shot,” into the cell. The particles are usually very fine gold pellets onto which the DNA has been stuck. With direct DNA absorption, a cell is bathed in the DNA, and an electric shock usually is applied (“electroporation”) to the cell to stimulate DNA uptake.

No matter what gene insertion method is used, a series of events must occur to allow a whole genetically modified plant to be recovered from the genetically modified cell: The cell must incorporate the new DNA into its own chromosomes, the transformed cell must initiate division, the new cells need to organize themselves into all the tissues and organs of a normal plant (“regeneration”), and finally, the inserted gene must continue to work properly (“**gene expression**”) in the regenerated plant.

To help ensure all this occurs, a “cassette” of genes is inserted during the initial transformation. In addition to the gene coding for the desired trait, other genes are added. Some of these genes promote the growth of only those plant cells that have successfully incorporated the inserted DNA. It might do this by providing the transformed cells with resistance to a normally toxic antibiotic that is added to the growth medium, for example. Other genes (“**promoters**”) may be added to control the functioning of the trait gene by directing when and where in the transformed plant it will operate.

The genes put into plants using genetic engineering can come from any organism. Most genes used in the genetic engineering of plants have come from bacteria. However, as scientists learn more about the genetic makeup of plants (“plant genomics”), more plant-derived genes will be used.

A “normal” melon (left) after five days, and a transgenic melon (right) after fifteen days. The ability of the modified fruit to be edible for a much longer period is a trait desirable for those in the produce business.

gene expression use of a gene to create the corresponding protein

promoters DNA sequences to which RNA polymerase binds to begin transcription

Agricultural Applications

Inserted genes can be classified into three groups based on their use: those that protect a crop, those that improve the quality of a harvested product, and those that let the plant perform some new function.

Genes That Protect a Crop. The major use of plant genetic engineering has been to make crops easier to grow by decreasing the impact of pests. Insect resistance has been achieved by transforming a crop using a Bt gene. Bt genes were isolated from *Bacillus thuringiensis*, a common soil bacterium. They code for proteins that severely disrupt the digestive system of insects. Thus an insect eating the leaf of a plant expressing a Bt gene stops eating and dies of starvation. There are many Bt genes, each of which targets a particular group of insects. Some Bt genes, for example, target caterpillars. Others target beetles.

Genetic engineering also has been used in the battle against weeds. Bacterial genes allow crops to either degrade herbicides or be resistant to them. The herbicides that are used are generally very effective, killing most plants. They are considered environmentally benign, degrading rapidly in the soil and having little impact on humans or other organisms. Thus a whole field of transgenic crops can be sprayed with broad-spectrum herbicides, killing all plants except the crops. Corn, soybeans, canola, and cotton that have been engineered to withstand either insects or herbicides, or both, are widely planted in some countries, including the United States. In addition, other crops, including potatoes, tomatoes, tropical fruits, and melons, have been engineered for resistance to viral diseases.

Genes That Improve Crop Quality. An emerging major use of genetic engineering for crops is to alter the quality of the crop. Fresh fruits and vegetables begin to deteriorate immediately after being harvested. Delaying or preventing this deterioration not only preserves a produce's flavor, and appearance, but maintains the nutritional value of the produce. Genes that change the hormonal status of the harvested crops are the major targets for genetic engineering toward longer shelf-life.

For example, the plant hormone ethylene is associated with accelerated ripening, as well as leaf and flower deterioration, in fruits that are injured or harvested. Scientists insert genes that interfere with a plant's ability to synthesize or respond to ethylene, thereby extending postharvest quality for many fresh products, including tomatoes, lettuce, and cut flowers. Scientists are also using gene insertion to improve a plant's nutritional value and color.

Genes That Introduce New Traits. One approach to improving the economic value of crops is to give them traits that are completely new for that plant. Some crops, including potatoes, tomatoes, and bananas, have been engineered with genes from **pathogenic** organisms. This is done to make animals, including humans, that eat the crops immune to the diseases caused by the pathogens. The genes code for proteins that act as **antigens** to induce immunity. Thus edible parts of plants are engineered to act as oral vaccines. This approach may be particularly effective for pathogens, such as those causing diarrhea and other gastrointestinal disorders, that enter the body through **mucous membranes**. This is because the "medicine" in the food comes into direct contact with these membranes and does not have to be absorbed into the blood stream. Genes have also been engineered into crop

pathogenic disease-causing

antigens foreign substances that provoke an immune response

mucous membranes nasal passages, gut lining, and other moist surfaces lining the body

plants to direct the plants to produce industrial enzymes used in the manufacture of paper. Other genes direct crops to produce small polymers useful in the manufacture of plastics. This general approach is being termed “plant molecular farming.”

Rice is another plant that has been engineered for a new trait. During commercial processing, a substantial part of the white rice grains are removed, leaving very little vitamin A. Vitamin A deficiency is a significant health problem in regions dependent on rice as a dietary staple. Scientists engineered a certain form of rice, known as “golden rice” because it has a yellow tinge, to express three introduced genes. These genes let the plant produce the **precursor** of vitamin A in the portion of the grain that remains after processing, thereby providing a dietary source of the vitamin. SEE ALSO AGRICULTURAL BIOTECHNOLOGY; BIOPESTICIDES; BIOTECHNOLOGY; CLONING GENES; CLONING ORGANISMS; GENE TARGETING; GENETICALLY MODIFIED FOODS; PLANT GENETIC ENGINEER; TRANSFORMATION.

Brent McCown

Translation

Translation is the cellular process in which the genetic information carried by the DNA is decoded, using an RNA intermediate, into proteins. This process is also known as protein synthesis.

Deciphering the Genetic Code

There are two steps in the path from genes to proteins. In the first step, called **transcription**, the region of the double-stranded DNA corresponding to a specific gene is copied into an RNA molecule, called messenger RNA (mRNA), by an enzyme called RNA polymerase. In the second step, called translation, the mRNA directs the assembly of amino acids in a specific sequence to form a chain of amino acids called a **polypeptide**. This process is accomplished by **ribosomes**, special amino acid-bearing RNA molecules called transfer RNAs (tRNAs), and other translation factors. The newly synthesized polypeptides form proteins, which have functional and structural roles in cells. All proteins are synthesized by this process.

The precise order of amino acids assembled during translation is determined by the order of **nucleotides** in the mRNA. These nucleotides are a direct copy of the linear sequence of the nucleotides in one of the two complementary DNA strands, which have been transcribed using a code in which every three bases of the RNA specify an amino acid. DNA and RNA molecules both have directionality, which is indicated by reference to either the 5' (“five prime”) end or the 3' (“three prime”) end.

The code is always read in the 5' to 3' direction, using adjacent, non-overlapping three-base units called codons. Since there are four different nucleotides (also called bases) in RNA (abbreviated A, C, G, and U), there are sixty-four (4^3) different codons, and each codon specifies a particular amino acid. There are only twenty different amino acids, however, so many of the amino acids can be specified by more than one codon, a circumstance that is known as degeneracy. The list of mRNA codons specific for a given

precursor a substance from which another is made

transcription messenger RNA formation from a DNA sequence

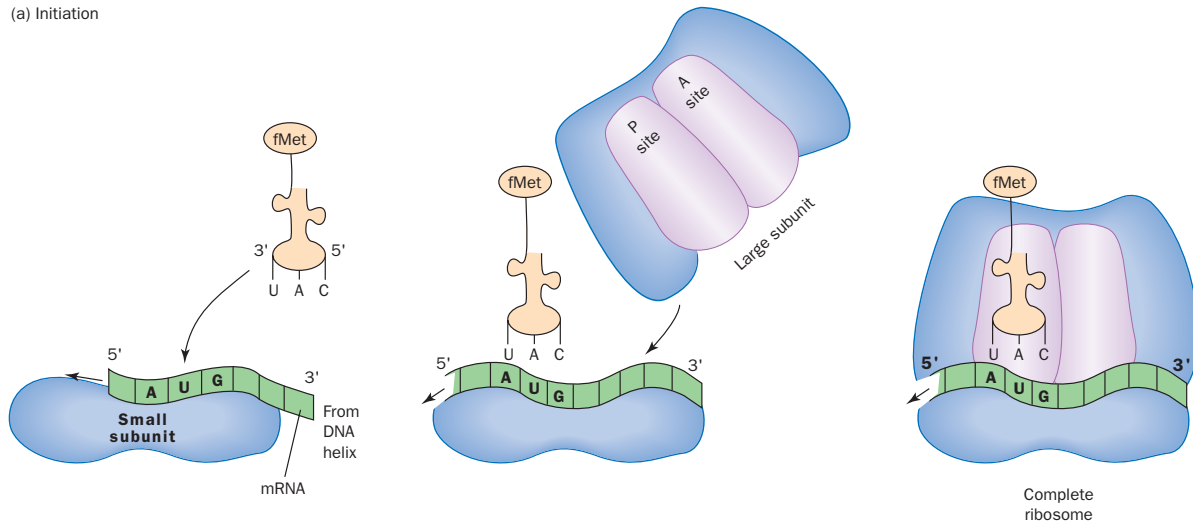
polypeptide chain of amino acids

ribosomes protein-RNA complexes at which protein synthesis occurs

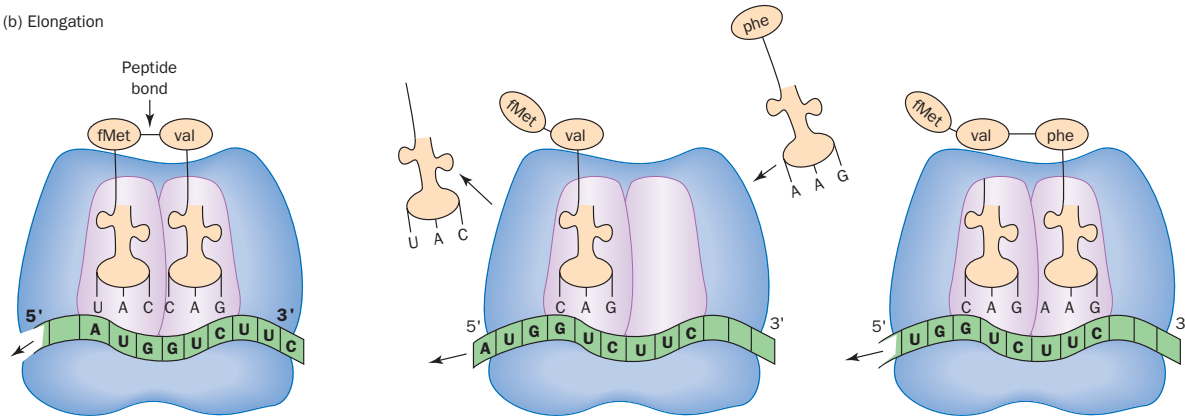
nucleotides the building blocks of RNA or DNA



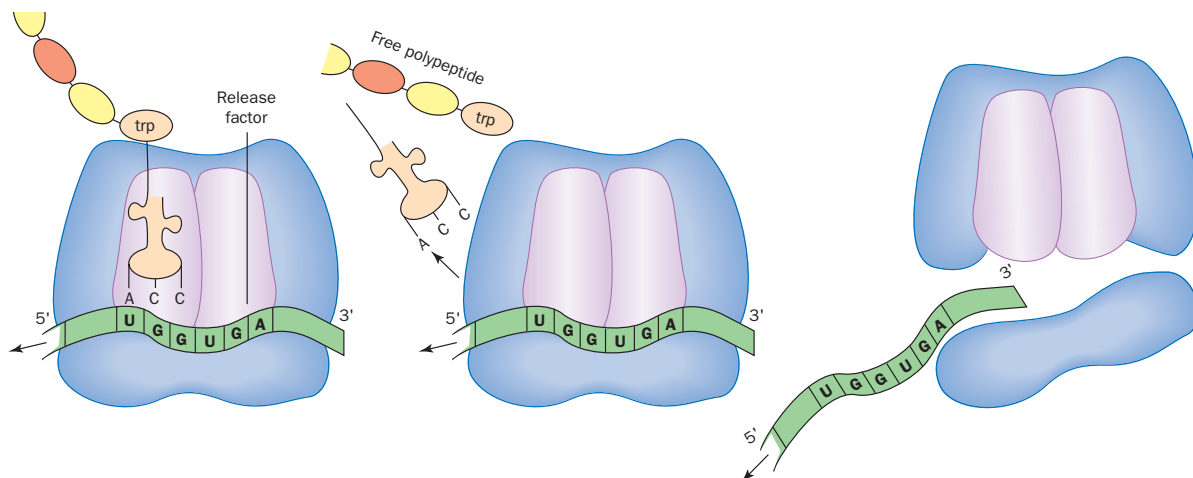
(a) Initiation



(b) Elongation



(c) Termination



The steps of translation. Initiation: RNA binds to the small ribosomal subunit, a tRNA binds to the RNA, and the large subunit attaches to the small subunit. Elongation: Successive tRNAs bind to the A site, form a peptide bond to the growing amino acid chain, and then move to the P site. The spent tRNA is ejected. Termination: A termination or stop codon binds a release factor, which dissociates two ribosomal subunits. The peptide (amino acid chain) is released. Adapted from Curtis and Barnes, 1994.

		Second base			
		U	C	A	G
First base	U	UUU } Phe	UCU } Ser	UAU } Tyr	UGU } Cys
		UUC }	UCC }	UAC }	UGC }
		UUA } Leu	UCA }	UAA } STOP	UGA } STOP
		UUG }	UCG }	UAG }	UGG } Trp
	C	CUU } Leu	CCU } Pro	CAU } His	CGU } Arg
		CUC }	CCC }	CAC }	CGC }
		CUA }	CCA }	CAA } Gln	CGA }
		CUG }	CCG }	CAG }	CGG }
	A	AUU } Ile	ACU } Thr	AAU } Asn	AGU } Ser
		AUC }	ACC }	AAC }	AGC } Arg
		AUA } Met	ACA }	AAA } Lys	AGA }
		AUG }	ACG }	AAG }	AGG }
	G	GUU } Val	GCU } Ala	GAU } Asp	GGU } Gly
		GUC }	GCC }	GAC }	GGC }
		GUA }	GCA }	GAA } Glu	GGA }
		GUG }	GCG }	GAG }	GGG }

Genetic code.

amino acid is called the **genetic code**. The start signal, or initiation codon, for translating the mRNA is usually specified by an AUG, which codes for the amino acid methionine. Three codons (UAA, UGA, and UAG) do not specify an amino acid. Instead, these codons serve as stop signals to indicate that the end of the gene has been reached. During the translation process, they signal that no further amino acids are to be assembled.

The process of translation is carried out by ribosomes, which bind the mRNA and conduct a **catalytic** activity, called **peptide bond** formation, for joining the amino acids. The amino acids are carried to the ribosome by the tRNAs. Each tRNA has a specific amino acid attached to it and contains a nucleotide triplet called an anticodon. The anticodon recognizes a specific codon on the mRNA by pairing with it, using base-pairing rules like those used by DNA: A pairs with U and G pairs with C. For example, a tRNA with a UUU anticodon recognizes the AAA codon. The amino acid lysine is attached to this tRNA, so every time the ribosome “reads” an AAA codon, the lysine-bearing tRNA is brought in, base pairs via its anticodon to the codon, and delivers a lysine to the growing protein chain.

Mutations arise when one or more bases in the DNA is changed. When the mutated DNA is transcribed, the resulting mRNA will carry the same mutation. Then, when the mRNA is translated, the amino acid sequence of the resulting protein will be different from the original, or **wild-type**, sequence because the codons affected by the mutation will recruit the wrong amino acids. The resulting mutant protein may have neutral, harmful, or even beneficial effects on the individual. These changes are the basis for evolution.

Stages of Translation

The process of translation can be broken down into three stages. The first stage is initiation. In this step, a special “initiator” tRNA carrying the amino acid methionine binds to a special site on the small subunit of the ribosome (the ribosome is composed of two subunits, the small subunit and the large

genetic code the relationship between RNA nucleotide triplets and the amino acids

catalytic describes substances that speed up a reaction without being consumed (e.g., enzyme)

peptide bond bond between two amino acids

wild-type most common form of a trait in a population

subunit). The mRNA is also loaded on, and positioned so that the initiation codon (usually AUG) is base paired with the anticodon of the initiator tRNA. The large subunit then binds to the small subunit. The resulting complex of ribosome, mRNA, and methionine-bearing initiator tRNA is called an initiation complex. Formation of this complex also requires a number of helper proteins called initiation factors.

The second stage is called chain elongation. During this stage, additional amino acids are progressively added. The methionine-bearing initiator tRNA sits on a site of the ribosome called the P (peptidyl) site. A new tRNA, bearing the next amino acid is base paired via its anticodon to the next codon of the mRNA, using a site called the A (acceptor) site. This new amino acid is then attached to the amino acid carried by the P site tRNA, forming a peptide bond. This enzymatic step is carried out by the ribosome, at a site called the peptidyl-transferase center.

The tRNA that has so far been attached to the amino acid in the P site is then released through the E (exit) site, and the new tRNA, now carrying both its own amino acid and the methionine moves into the P site. The mRNA also slides three bases to bring the next codon into position at the A site. A third tRNA, again carrying a specific amino acid and recognizing the third codon of the mRNA, moves into the A site, and the cycle is repeated. As these steps are continued, the mRNA slides along the ribosome, three bases at a time, and the peptide (amino acid) chain continues to grow. As with initiation, elongation requires helper proteins, called elongation factors. Energy is also required for peptide bond formation.

The final stage of translation is termination. The signal to stop adding amino acids to the polypeptide is a stop codon (UAA, UAG, or UGA), for which there is no partner tRNA. Rather, special proteins called release factors bind to the A site of the ribosome and trigger an enzymatic reaction by the ribosome. This reaction causes the ribosome to release the polypeptide and mRNA, ending the elongation process.

At a given time, more than one ribosome may be translating a single mRNA molecule. The resulting clusters of ribosomes, which resemble beads on a string, are called polysomes.

Recognition of Initiation Codons

Not all AUG codons serve as the site of initiation. Most AUGs are intended to code for methionines within the polypeptide chain. Therefore, in addition to the methionine-bearing initiator tRNA, another set of methionine-specific tRNAs are used for these internal AUG codons. The ribosome must be able to distinguish between these two kinds of AUG codons. In bacteria, additional information contained within the mRNA sequence immediately before the intended initiating AUG, called a Shine-Dalgarno sequence, helps the ribosome to recognize where it should start translating. Any AUG sequences on the 5' side of the initiation codon are ignored. In eukaryotic cells, a different strategy is used to recognize the initiating AUG codon. The mRNA contains a special structure at its 5' end, which helps the ribosome to attach and then to scan down the RNA molecule until it reaches the first AUG triplet. In bacteria and eukaryotes, AUG codons encountered during translation after initiation are recognized by a non-initiator methionine-bearing

tRNA. SEE ALSO GENETIC CODE; MUTATION; NUCLEOTIDE; PROTEINS; READING FRAME; RIBOSOME; RNA; TRANSCRIPTION.

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Transplantation

Modern medicine continues to offer many miracles that lengthen the life spans of humans, as well as greatly increase the quality of life that they enjoy. If one were to draw up a “top ten” list of technical feats, surely the ability to successfully transplant an entire organ from one human to another would be high on the list. Transplantation can be defined as the transfer of cells, tissues, or organs from one site in an individual to another, or between two individuals. In the latter case, the individual who provides the transplant organ is termed a donor, and the individual receiving the transplant is known as the recipient. The science of transplant biology has, in fact, become a victim of its own success, in that the demand for organs exceeds the supply of donors.

Types of Transplants

There are four basic types of transplants, which reflect the genetic relationship of the recipient to the donor. The autograft is the transfer of tissue from one location of an individual’s body to another location that is in need of healthy tissue; in other words, the recipient is also the donor. Common examples of autografts are skin transplants in burn patients and bypass surgery in patients suffering from coronary heart disease. The syngraft is a transplantation procedure carried out between two genetically identical individuals. These types of transplants, like autografts, are always successful, unless there have been technical problems during the surgery. The first successful human kidney transplant was a syngraft, carried out in 1954 between identical twins.

An allograft is the transfer of tissue or an organ between nonidentical members of the same species. This is the predominant form of transplantation today, and allografts have dominated transplant research for many years. Finally, the xenograft represents the most disparate of genetic relationships, because it is the transfer of tissue or organs between members of different species. Many think that xenografts are the answer for solving the shortage of transplant tissue and organs that we are currently experiencing. Both allografts and xenografts have the disadvantage that the recipient’s immune system is designed to recognize and reject foreign tissue.

The Genetic Basis of Transplant Rejection

Research that began in the 1940s gave geneticists the first hints that a portion of the mammalian **genome** contained a cassette of genes that governed the acceptance or rejection of transplanted tissues. This grouping of genes was labeled the major histocompatibility complex (MHC). Subsequently, it has been found that the MHC also contains genes that are

genome the total genetic material in a cell or organism



Five “knockout” piglets, considered to be the first of their kind, were raised with the intention of using their organs for human transplants. Knockout pigs have the gene inactivated that usually leads human immune systems to rejecting pig organs.



involved in governing antibody responses as well. MHC molecules are identical between identical twins, but are otherwise different for every individual. Thus they allow the body to distinguish “self” from “nonself” on the molecular level.

antigen a foreign substance that provokes an immune response

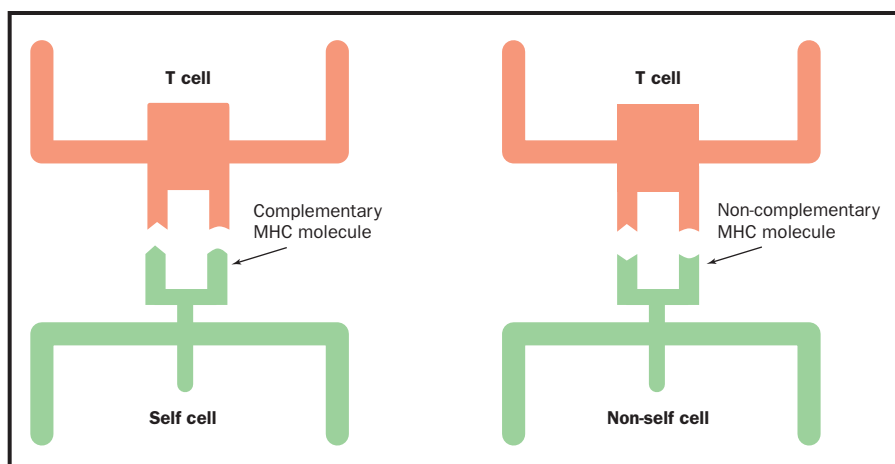
The immunogenicity (ability to induce an immune response) of major transplantation **antigens** is so strong that differences between the antigens of the donor and recipient is enough to trigger an acute rejection response. To the extent that it is possible, therefore, the recipient and donor are matched for MHC type, to minimize acute rejection.

However, there are cases in which the donor and recipient are very well matched, and yet rejection of the graft still occurs. This is due to other genes found in various places in the genome, known as minor histocompatibility genes, that encode for other weaker transplantation antigens, or foreign peptides, that can cause a chronic rejection response. Currently, researchers have not been able to determine the extent or location of all of these genes. Results obtained from the mapping of genes in the human genome will aid in overcoming this problem.

The Mechanisms of Transplant Rejection

The immune system’s attack on foreign tissue is mediated by lymphocytes, phagocytic cells, and various other white blood cells. Various subgroups of lymphocytes have different responsibilities. Once stimulated, the B-lymphocytes (derived from bone marrow) will develop into a cell that produces antibodies (soluble proteins that specifically seek out invaders). Antibodies may cause hemorrhaging by attaching to the lining of blood vessels in the transplant and then activating a naturally occurring series of potent enzymes known as the complement system.

The T-lymphocyte (derived from the thymus) can develop either into a T-helper cell, which serves a regulatory function, or a T-cytotoxic (killer) cell. Activated T-helper cells induce T-cytotoxic cells to destroy a foreign graft by attacking those cells in the transplant that display incompatible antigens.



When a T cell detects a noncomplementary MHC molecule on a cell, it interprets it as “nonself” and sets a rejection reaction in motion.

The task of the transplant treatment team is to somehow derail this natural process of reacting to foreign tissue long enough for the graft to “heal in” and survive without at the same time putting the patient at risk for increased infectious disease. To control this type of response, various immunosuppressive drugs, such as cyclosporine, have been developed. Great strides have been made in controlling rejection of transplanted tissue and organs by these methods.

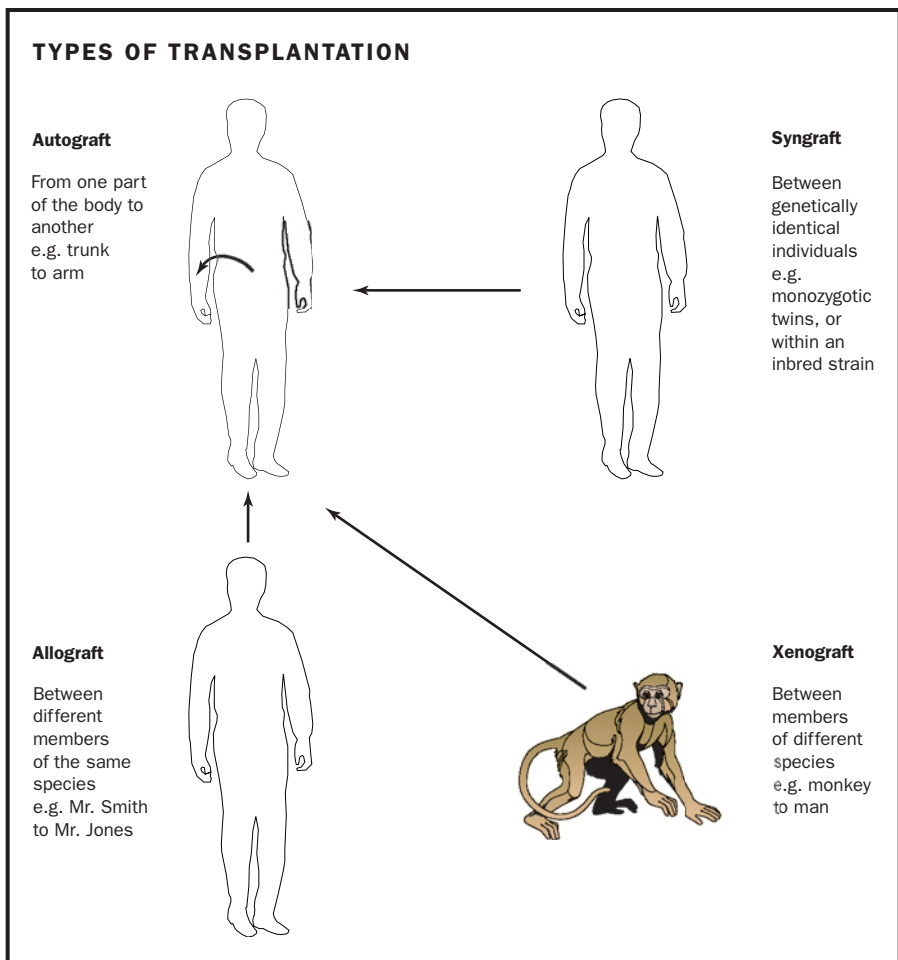
The Supply Crisis in Transplantation

The predominant issue in transplantation biology is now one of increasing the supply of organs for patients in need of them. This is not only a technical problem, but in some cases, raises ethical issues as well. For instance, there have been cases of parents with a sick child purposely conceiving a second child for the main purpose of being a bone marrow donor for their ailing offspring. There has also been the rise of a black market in body parts, particularly emanating from China, in which various organs from executed prisoners are offered for sale.

Researchers have come up with numerous new options to improve on the availability of organs needed for transplantation. For instance, chemicals can be used to stimulate a patient’s own stem cells (cells that can develop into almost any type of tissue, depending upon the local influences it encounters) to migrate from the bone marrow to the diseased organ, develop into the right type of cell, and regenerate the organ. A more controversial application of stem cell research involves the use of embryonic stem cells. One version of this strategy is to remove DNA from the patient’s own skin cells, inject it into a donated human egg from which the nucleus has been removed, and then allow that egg to develop into an early-stage embryo. The embryo can then be harvested for embryonic stem cells that can be influenced into growing into the organ of choice. Another major strategy is to collect embryonic stem cells from aborted fetuses or from umbilical cord blood. This whole topic has become a very highly debated issue due to the involvement of human embryos, as has the entire burgeoning field of stem cell-applied medical treatment.



Adapted from Roitt, 2001.



Another option is called tissue scaffolding. The strategy in this technique is to take the patient's cells (or cells from a donor) and inject them into a three-dimensional scaffold of biodegradable polymers that are in the shape of the desired organ. The entire structure is transplanted, and the cells in the scaffold replicate, reorganize, and form a new organ. Research indicates that cells are surprisingly adept at regenerating the tissue of their origin. As the cells grow, and the polymers of the scaffold naturally degrade, what is left is a new, functioning organ. The obvious problem with this technique, as with options using stem cells, is that it takes time to form the new organ. This is of little help to a patient who is in immediate need of a transplant.

One of the most promising, and controversial, sources for new organs for humans are xenotransplants from other species, particularly baboons and pigs. Many individuals are very strongly opposed to raising animals for the sole purpose of harvesting their organs for humans, viewing it as inhumane. Another area of controversy, particularly concerning baboon donors, is the possibility of spreading unknown diseases into the human population. There are already established precedents for viral diseases jumping from primates to humans, such as the AIDS virus (HIV), Ebola virus, and the hantavirus. Consequently, there is a fear that xenotransplantation could unleash a new plague upon humans. More and more xenotransplant research is moving toward the use of pigs, since it is very much less likely that a pig virus could

infect a human. The development of **pathogen**-free colonies of pigs would also greatly reduce the likelihood of such an occurrence.

The real advantage to using pigs is that they are easily bred, mature quickly, and their organs are of a comparable size to that of humans. In addition, pigs are amenable to genetic engineering, whereby the genes that encode transplantation antigens that would be recognized by a human recipient could be removed so that the resulting organs would not be recognized as foreign in the human. In addition, pigs have now been cloned, so that once such an antigen-free animal has been constructed, we could have a continuous source of immunologically nonstimulating organs available for transplantation into human patients. SEE ALSO AGRICULTURAL BIOTECHNOLOGY; CLONING ORGANISMS; EMBRYONIC STEM CELLS; IMMUNE SYSTEM GENETICS.

Richard D. Karp

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Transposable Genetic Elements

Transposable genetic elements (TEs) are segments of DNA that can be integrated into new chromosomal (genomic) locations either through direct DNA transfer (transposons), or via an RNA intermediate (retrotransposons). Pseudonyms for TEs include mobile elements, jumping genes, genomic parasites, and selfish DNA. TEs are known to be responsible for several human genetic diseases and may play a role in evolution in many species.

Early Evidence

Barbara McClintock originally theorized that unusual patterns of phenotypic variance (in corn kernels) could be explained by gene transposition. However, this explanation did not coincide with traditional Mendelian inheritance that genetic information was fixed within the **genome**, and her views were not widely accepted within the scientific community until the 1960s, when evidence of transposition began to accumulate.

McClintock was a cytogeneticist working on maize. She noted that, in maize, there was a pigment-bearing layer of the kernel, called the aleurone layer, that changed color from kernel to kernel and generation to generation. She also noted a baffling result that occurred when a **homozygous** plant for purple aleurone (CC) was crossed with a colorless aleurone homozygote (cc). About one-half of the kernels of offspring corn were solid purple, and one-half were purple with varying sizes of colorless spots, suggesting breakage (and loss) of the C locus. However, one of 4,000 analyzed

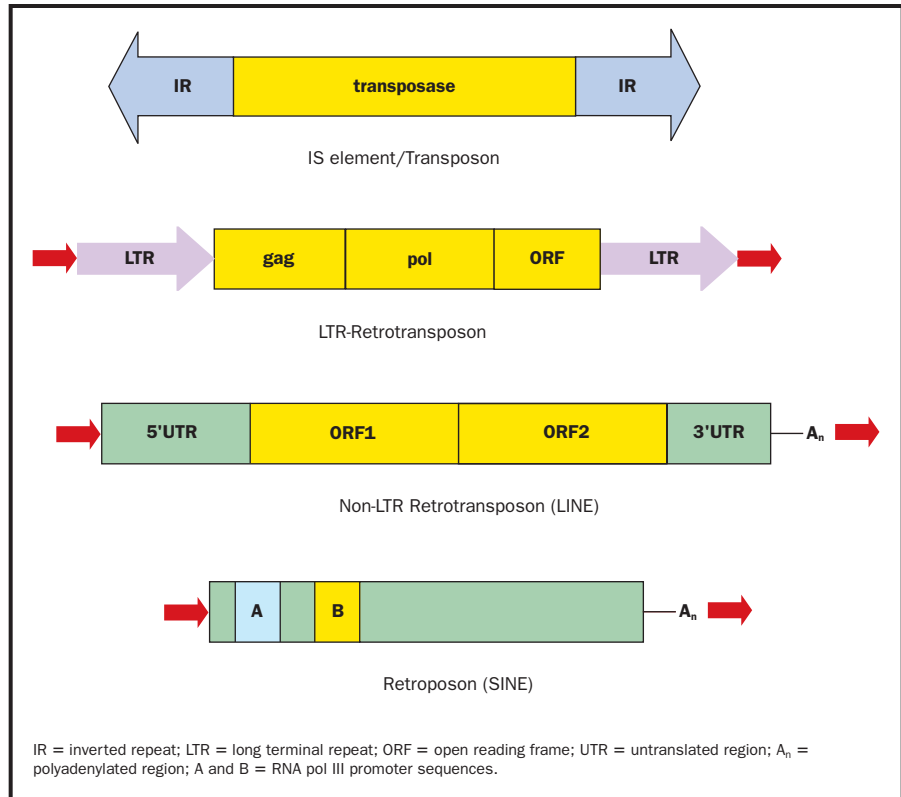
pathogen disease-causing organism

genome the total genetic material in a cell or organism

homozygous containing two identical copies of a particular gene



Figure 1. Schematic diagrams of various classes of transposable elements.



locus site on a chromosome (plural, loci)

kernels was colorless with purple spots, indicating a “gain-of-function” of the *c* locus. McClintock identified a **locus** (called *Ds*, which stands for dissociation) that induced nearby chromosome breakage only in the presence of another gene, which she called *Ac* (for activator). She concluded that the inactivation of pigment production is caused by insertion of the *Ds* gene into the *C* locus (disrupting pigment products and yielding the colorless background), and the eventual movement of the *Ds* out of the gene, restoring pigment production and yielding the purple spots. All of this could occur within a single kernel.

Further support for the mobility of the *Ds* and *Ac* genes was the inability to determine their chromosomal locations, which differed among plants. High mutational reversion rates have been subsequently identified in other organisms, with mobile elements now offering a plausible explanation.

TEs across the Evolutionary Tree

TEs are ubiquitous throughout the evolutionary tree from microorganisms to mammals. For instance, a bacterial virus called phage MuM (mutator) is characterized as a TE based on similarities with other transposons. Mu integration into the host bacterial chromosome is considered transposition because it can occur nearly anywhere, thereby inactivating host genes and generating insertions and deletions.

Transposons also occur naturally in bacterial genomes. The extensively studied species *Escherichia coli* (*E. coli*) contains insertion sequences (IS) and transposons in its genome. IS elements are generally small, have inverted sequence repeats at their ends (important for their mobility), and contain

Transposase is an enzyme that catalyzes transposition. It may be encoded in the transposon or may reside elsewhere.

an overlapping genetic region encoding a transposase and a repressor. Transposons are larger than IS elements, since they contain additional genes such as drug-resistance genes. These elements are flanked either by inverted repeats or by IS elements. Bacterial transposons undergo conservative transposition, in which the transposon is excised and pasted elsewhere, or replicative transposition, in which it is copied and the copy is inserted elsewhere.

Ty elements in yeast contain retroviral-related sequences (called *gag* and *pol*) and include long terminal repeats (LTRs); hence they are considered viral (or viral-like) retrotransposons or LTR retrotransposons. Their activity is replicative: An RNA is transcribed from the gene, reverse-transcribed to DNA (cDNA) and integrated elsewhere in the genome. Since Ty does not contain *env* genes, which code for encapsulating envelope proteins, it does not yield infectious particles. However, viruslike particles accumulate in cells in which retroposition has been induced.

The African Trypanosome, a parasitic protozoan, contains the *ingi* (non-LTR) retrotransposons. *Ingi* may therefore be referred to as a parasite's parasite. Full-length elements are 5.2 kilobase pairs long, have multiple adenine nucleotides at one end (called a "poly A tail"), and DNA sequences similar to reverse transcriptase genes and mammalian LINES (discussed below). Among insects, the *Drosophila* genome contains a virtual cornucopia of TEs. Fruit fly transposons include mariner, hobo, and *P* elements, non-LTR-retrotransposons include *I*, *F*, and *jockey* elements, and LTR-retrotransposons such as gypsyM and copia-like elements. These eukaryotic transposons are similar to bacterial IS elements, but are generally larger due to the presence of introns (noncoding sequences of the genome).

The primary TEs in mammalian genomes include short and long interspersed repetitive elements (SINEs and LINES, respectively). SINEs represent a group of small retrotransposons (75–500 **base pairs**) and lack protein-coding sequences. In primates, *Alu* elements represent the predominant SINE family. More than 1 million copies of *Alu* are contained in the human genome, representing about 13 percent of the genome. This is truly impressive considering their lack of replicative autonomy. *Alu* elements are 300 base pairs in length, are rich in adenine sequences both internally and at the 3' (downstream) end, and internal RNA polymerase III promoter sequences, allowing them to be replicated. L1 elements represent the predominant primate LINE family, contain two **open reading frames**, adenine-rich 3' ends, and internal RNA polymerase II promoter sequences. They constitute approximately 20 percent of the human genome. A full-length LINE is about 6.5 kilobase pairs long, although most elements are truncated as a result of incomplete reverse transcription.

Transmission of TEs

TEs generally demonstrate vertical transmission, meaning that new incorporations are inheritable by offspring. Although transposons may excise from their genomic location and integrate elsewhere, retrotransposons form stable integrations, creating a molecular "fossil record" of past integration events.

Over 99 percent of human *Alu* elements are shared with the chimpanzee genome. Unlike *Alu*, L1 predates the origin of primates, with many

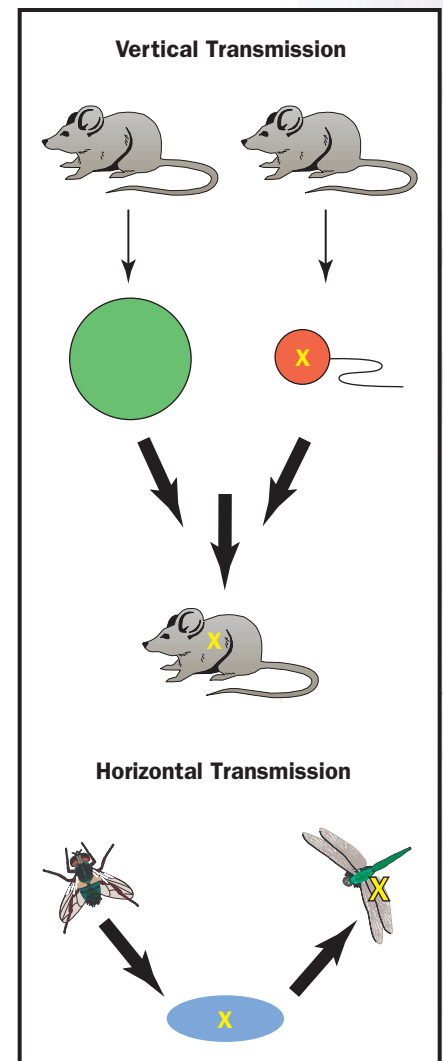
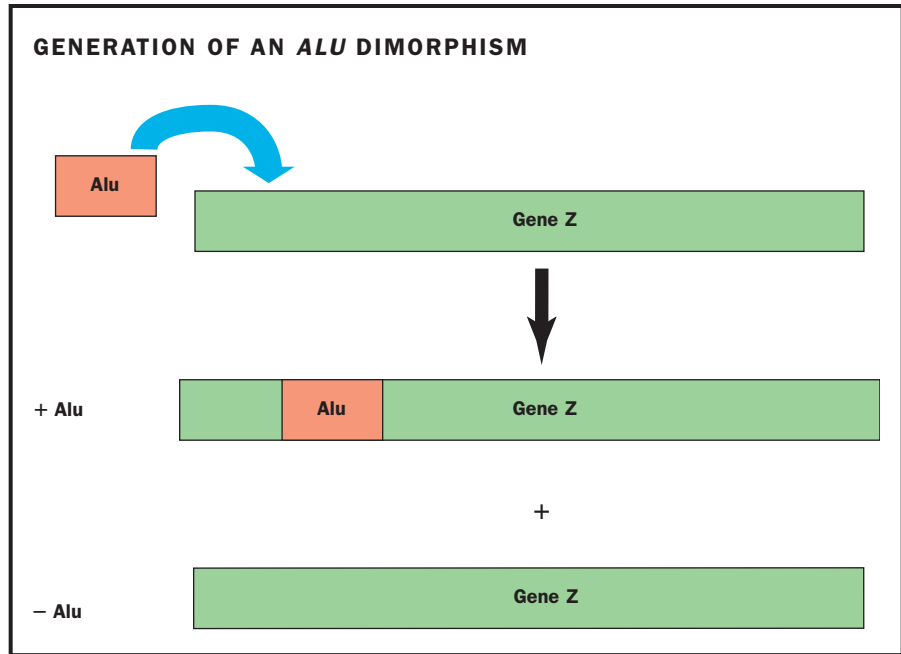


Figure 2. Transmission of transposable elements (TEs). Vertical transmission involves a germ-line TE integration (indicated by the X in the sperm), that is passed on to the next generation. Horizontal transmission involves the transfer of a TE (indicated by the X) across species, possibly via a parasite.

base pairs two nucleotides (either DNA or RNA) linked by weak bonds

open reading frames DNA sequences that can be translated into mRNA; from start

Figure 3. Creation of presence/absence variants (dimorphism) by the integration of an Alu element. The Alu increases the gene size by 300 bp.



integrations predating the origin of placental mammals. Most recent Alu and L1 integrations in the human genome generate insertion presence/absence variants that are referred to as dimorphisms.

Dimorphisms provide DNA markers useful in mapping studies, fingerprinting, and human population investigations. Retrotransposon integrations are also associated with human disorders. Examples include Alu integrations into NF1, factor IX, and BRCA2 genes yielding neurofibromatosis, hemophilia, and breast cancer, respectively, and L1 integrations into the *c-myc*, APC, and factor VIII genes cause breast cancer, colon cancer, and hemophilia, respectively.

Horizontal transfer of TEs, or the transmission between species, has also been implicated. General evidence involves the lack of **phylogenetic** correspondence between the TEs and their host organisms. For example, outside of the *Drosophila melanogaster* species group, *jockey* has been detected only in the distantly related *D. funebris*, suggesting that it was transferred between the two. *P* elements have also exhibited horizontal transfer, possibly from *D. willistoni* to *D. melanogaster*, but more importantly, strains of *D. melanogaster* are lacking the element, indicating recent spreading through populations. Additionally, insect-related mariner elements, characterized primarily by their transposase, have been identified in diverse organisms such as flatworms and hydra, yet are lacking in twenty other invertebrate species representing major phyla. These transfers are both ancient and relatively recent, as one element has 92 percent amino acid similarity between Hydra and a staphylinid beetle. Possible transmission **vectors** include parasites (such as mites) and viruses. Some evidence also exists for horizontal transfer of mammalian TEs, including the putative discovery of an Alu element in the malarial parasite *Plasmodium vivax*.

phylogenetic related to the evolutionary development of a species

vectors carriers

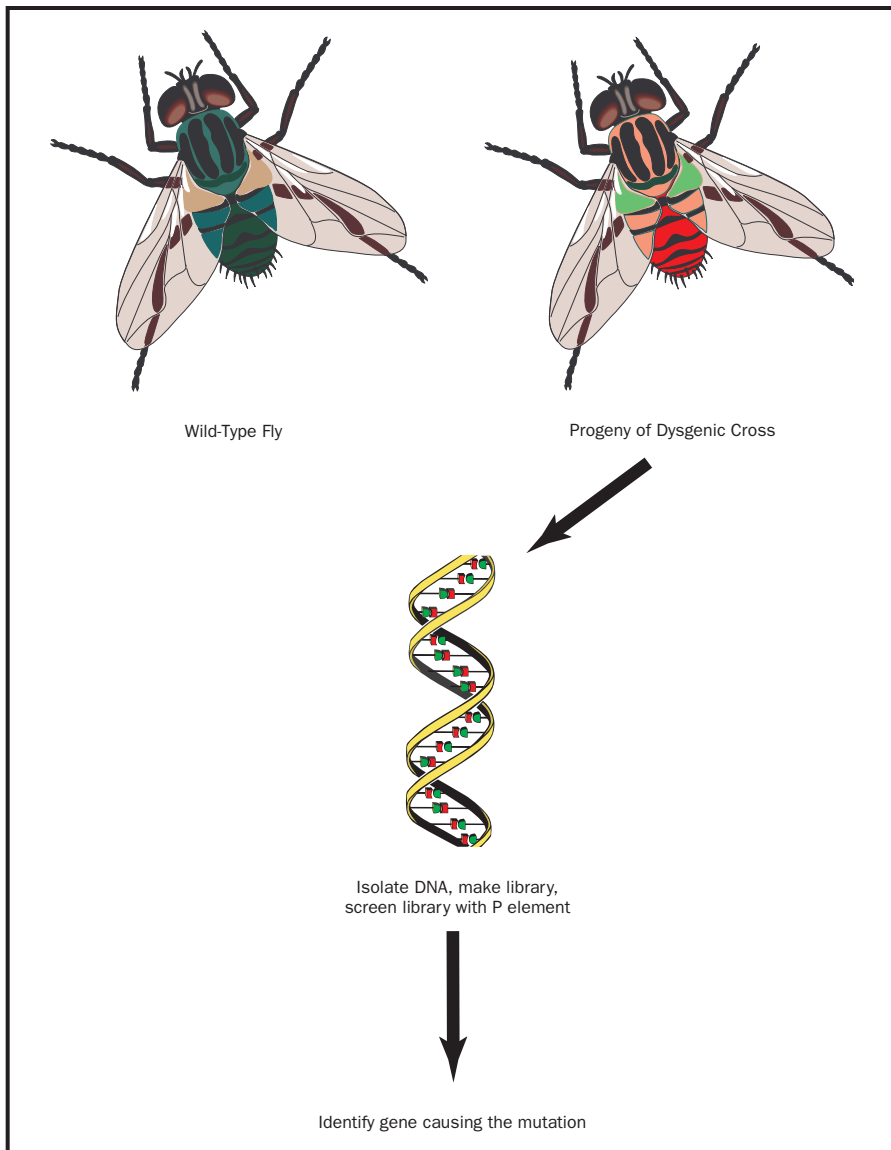


Figure 4. Identifying genes using transposable elements (TE). A dysgenic fly with a randomly incorporated mutation is generated. DNA is isolated to produce a library which is screened with the TE to identify the mutated gene.

Transposition Mechanisms

Transposon mobility may be either nonreplicative or replicative. In non-replicative mobility, the TE is cut out of its original position and integrated at a new location. In replicative mobility, transposition makes staggered cuts in donor and recipient sites, followed by a complex transfer, replication, and resealing operation.

SINE and LINE generation originates from “master genes”: Only a few of the thousands to millions of copies are capable of serving as the source for new integrating elements. The proposed details of copy formation and integration differ between the two types of generation.

TEs and Species Evolution

Are TEs simply genomic parasites, or have they had a major impact on the evolution of species? Certainly, a TE integration could have an immediate



detrimental impact on its host. However, specific and cumulative integrations may provide a mechanism for speciation. *Drosophila* *P* elements have been implicated in generating a reproductive barrier between strains. When females devoid of active elements are crossed with males of a strain with active elements, the TEs run rampant in developing germ-line cells, yielding various chromosomal anomalies and F_1 hybrid sterility. This may be an important mechanism in promoting the creation of new species from those strains.

Arguably, SINEs and LINEs may also drive species evolution through several mechanisms. They can disrupt or reset coordinated gene regulation, facilitate the pairing of homologous chromosomes, and possibly offer sites for genomic imprinting. In addition to individual integrations, LINEs have contributed to genomic diversity by delivering adjacent sequences, including whole genes, to new genomic locations. TEs also offer numerous sites for homologous unequal recombination.

Transposons as Molecular Biology Tools

Transposons can be used to facilitate cloning of genes, identify regulatory elements, and produce transgenic organisms. For example, transposon tagging involves inducing transposition of a TE, allowing for disruption of a gene that generates an organism with a mutant phenotype, and is followed by molecular techniques that allow for the identification of the gene. A variation of transposon tagging (enhancer trapping) uses *P* elements to identify DNA sequences that regulate the expression of genes. *P* elements can also be used to incorporate foreign genes into fruit flies (transgenics). In addition, transposon fossils have been useful for the isolation of species-specific DNA from complex sources such as using inter-Alu PCR for the isolation of human genomic DNA sequences. SEE ALSO DNA LIBRARIES; EVOLUTION OF GENES; IMPRINTING; MCCLINTOCK, BARBARA; REPETITIVE DNA ELEMENTS; RETROVIRUS; REVERSE TRANSCRIPTASE; YEAST.

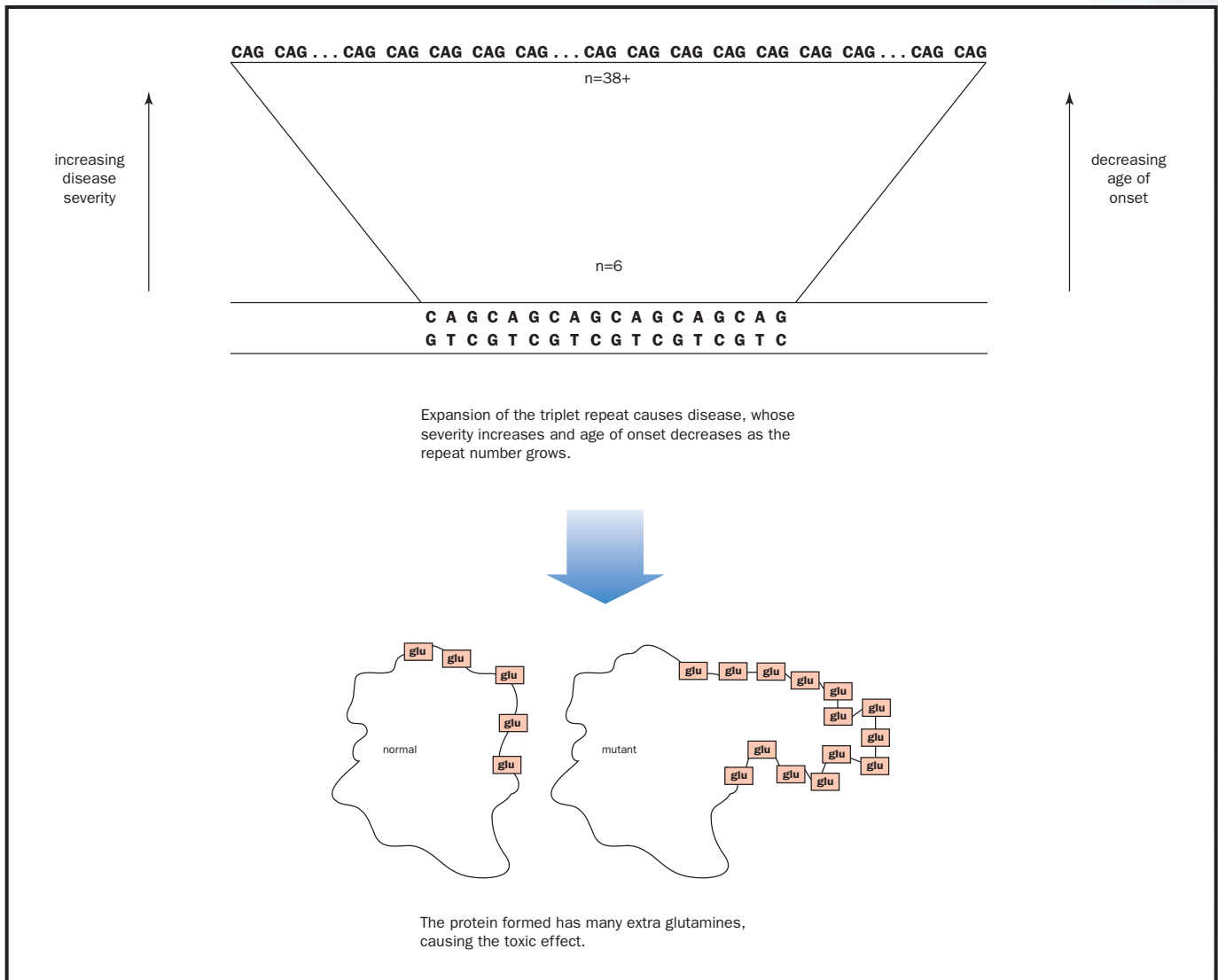
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Triplet Repeat Disease

Trinucleotide, or triplet repeats, consist of three consecutive nucleotides that are repeated within a region of DNA (for example, CCG CCG CCG



CCG CCG). These are found in the **genome** of humans and many other species. All possible combinations of nucleotides are known to exist as triplet repeats, though some, including CGG and CAG, are more common than others. The repeats may be within genes or between genes. In genes, they may be in regions that specify proteins (coding regions called exons) or in noncoding regions (introns). If present within exons, they may be present in translated regions and hence encode a series of identical amino acids, or they may occur in regions not translated into protein. Triplet repeats are frequently found in genes that encode transcription factors and those involved in regulating development.

Implicated in Diseases

Triplet repeats were once thought to be apparently benign stretches of DNA. However, it is now known that these repeats can sometimes undergo dynamic or expansion mutation. In this type of mutation, through mechanisms during DNA replication that are only partly understood, the number of triplets in a repeat increases (expands) and can cause disease.

Expansion of a triplet repeat gene segment leads to increased disease severity and decreased age of onset in proportion to the degree of expansion. Repeats that are translated into protein alter the structure of the protein, conferring new functions.

genome the total genetic material in a cell or organism

somatic nonreproductive; not an egg or sperm

germ cells cells creating eggs or sperm

Diseases with Triplet Repeats in Noncoding Regions

Fragile X syndrome (CGG repeat)
Fragile XE syndrome (GCC repeat)
Friedreich ataxia (GAA repeat)
Myotonic dystrophy (CTG repeat)
Spinocerebellar ataxia type 8 (CTG repeat)
Spinocerebellar ataxia type 12 (CAG repeat)

Diseases with (CAG)_n Repeats in Coding Regions

Spinobulbar muscular atrophy (Kennedy's disease)
Huntington's disease
Dentatorubral-pallidoluysian atrophy
Spinocerebellar ataxia types 1, 2, 3, 6 and 7

pathogenesis pathway leading to disease

phenotypes observable characteristics of an organism

kinase an enzyme that adds a phosphate group to another molecule, usually a protein

There are many diseases known to be caused by triplet repeats. They share certain common features. The mutant repeat length is unstable in both **somatic** and **germ cells** of the body, meaning it can change in length during DNA replication. Also, the triplet repeat often expands rather than contracts in successive generations. Increasing repeat size is correlated with decreasing age of onset or increasing disease severity in successive generations. This phenomenon is called anticipation, and is a characteristic of most triplet repeat diseases.

Classification of Triplet Repeat Disorders

Triplet repeat disorders fall into two subclasses, depending on the location of the trinucleotide repeat within the gene. The first subclass has triplet repeats occurring in noncoding sequences of DNA, either introns or regions at the start or end of the gene, called 5' ("five prime") or 3' ("three prime") untranslated regions. The second subclass is characterized by (CAG)_n repeats that code for repeated stretches of the amino acid glutamine (polyglutamine) in coding regions of the affected gene.

Noncoding Triplet Repeat Disorders

The noncoding triplet repeat diseases typically have large and variable repeat expansions that result in multiple tissue dysfunction or degeneration. The triplet repeat sequences vary in this subclass (CGG, GCC, GAA, CTG, and CAG). It is clear that the particular triplet sequence and its location with respect to a gene are important defining factors in dictating the unique mechanism of **pathogenesis** for each disease. The pathogenic mechanism also varies from disease to disease depending on the consequences of the lost function of the respective proteins or in some cases acquired function of a toxic transcript.

Fragile X Syndrome. This disorder is caused by the expansion of a (CGG)_n repeat in the 5' untranslated region of the fragile X mental retardation (*FMR1*) gene. Normally there are about six to fifty-three repeats in this region. In the disease state there are more than 230 triplet repeats. This causes the transcriptional silencing of the gene, leading to loss of normal gene function. Some of the symptoms seen in this disorder include mental retardation, deformed features, and hyperactivity.

Myotonic Dystrophy (DM). DM is a multisystem disorder with highly variable **phenotypes** and anticipation. Rigidity of muscles after contraction (tonic spasms), muscle weakness, and progressive muscle wasting characterize adult-onset DM. Developmental abnormalities, mental handicap, and respiratory distress are often evident in more severe forms. DM is caused by an expanded CTG triplet repeat in the 3' untranslated region of the protein **kinase** gene *DMPK*. The CTG expansion may disrupt *DMPK* transcription, causing loss of function. CTG expansions may also cause loss of function in two genes flanking *DMPK*: the DM locus-associated homeodomain protein (DMAHP) and gene 59 (also known as *DMWD*). The CTG expanded transcript could also gain toxic function by interfering with normal processing of various RNAs. Congenital myotonic dystrophy occurs from birth in infants whose mothers have DM, often a case so mild it is never diagnosed. Due to anticipation, the child is much more severely affected than the mother.

Polyglutamine Diseases

In comparison with the noncoding disorders, polyglutamine diseases have triplet (CAG) repeat expansions in coding regions of genes. Although the mutant proteins do not share any homology (similar sequences) outside the polyglutamine tract, the polyglutamine diseases have several similar features and may share common mechanisms of pathogenesis.

A simple loss of normal function of the gene does not account for the phenotype seen in these diseases. Studies of animal models and tissue culture systems have demonstrated that the mutant protein is toxic. The mutant protein can aggregate and form inclusions in the cytoplasm and nucleus. Also, all the polyglutamine diseases known so far are characterized by progressive neuronal dysfunction that typically begins in midlife and results in severe neurodegeneration. Despite the ubiquitous expression of the mutant genes, only a specific subset of neurons is vulnerable to neurodegeneration in each disease. What makes these specific neurons more susceptible than other neurons and other cells of the body remains a mystery.

Huntington's disease is caused by a CAG expansion in the *HD* gene that codes for a protein called huntingtin. This protein has no known function. In the normal, nonmutant form of the gene there are about six to thirty-five CAG repeats. In the disease-causing **allele** there are usually more than thirty-seven CAG repeats. Patients typically show symptoms of dementia, involuntary movements, and abnormal posture.

Spinobulbar muscular atrophy (Kennedy's disease) is caused by CAG expansion in the *AR* gene that codes for the **androgen** receptor protein. The normal gene contains nine to thirty-six CAG repeats and the mutant gene usually contains thirty-eight to sixty-two repeats. This disease of motor neurons (nerve cells controlling muscle movement) is characterized by progressive muscle weakness and atrophy. Over 50 percent of the affected males may also have reduced fertility.

Why Are Polyglutamines Toxic?

Several hypotheses have been offered to explain the toxic nature of polyglutamine repeats, though none has yet been conclusively proven, and may yet be wrong in either details or central concept. The Nobel Prize recipient Max Perutz suggested that the expanded polyglutamine repeats promote the formation of protein aggregates. These aggregates often contain ubiquitin, a marker for protein degradation. Expanded polyglutamine proteins may adopt energetically stable structures that resist unfolding and therefore impede clearance by the cell's protein-degrading machinery, called the proteasome. This concept is supported by the observation that addition of proteasome inhibitors promotes aggregation of mutant huntingtin, ataxin-1, and ataxin-3 in cell culture.

Most polyglutamine aggregates are found in the nucleus. Is nuclear localization of the mutant proteins a critical event in polyglutamine disease, and if so, how do the mutant proteins affect nuclear function? Aggregates may interfere with such important events occurring in the nucleus as gene transcription, RNA processing, and nuclear protein turnover. The aggregates formed may sequester and deplete critical nuclear factors

allele a particular form of a gene

androgen testosterone or other masculinizing hormone



required for transcription. Recent evidence suggests that mutant huntingtin does interfere with transcription factors.

Scientists have only begun to understand the role of triplet repeats in disease. The cloning of the disease gene and the identification of expanding repeats represent preliminary steps to the understanding of the full disease process. Genetic testing provides at least an immediate diagnostic tool, but much still remains to be determined for effective therapies to be developed for tomorrow's patients. Many questions, including the biological role of these triplet expansions in evolution, remain unanswered. SEE ALSO ANDROGEN INSENSITIVITY SYNDROME; FRAGILE X SYNDROME; GENE; GENETIC TESTING; INHERITANCE PATTERNS; PLEIOTROPY.

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Internet Resources

- Huntington's Disease Society of America. <www.hdsa.org/>.
- International Myotonic Dystrophy Organization. <<http://www.myotonicdystrophy.org/>>.

Trisomy See *Chromosomal Aberrations*

Tumor Suppressor Genes

Tumor suppressor genes regulate **mitosis** and cell division. When their function is impaired, the result is a high rate of uncontrolled cell growth or cancer. Damage to tumor suppressor genes contributes to a large number of different types of **tumors**.

The Balancing Act of Regulating the Cell Cycle

The **cell cycle** is a fundamental process of life, regulated by a balance of positive and negative mechanisms that act at key points during cell growth and differentiation. Proto-oncogenes tend to "push" the cell cycle (in a positively acting manner) by activating various cell cycle pathways within the developing cell. By contrast, tumor suppressor genes normally repress, or "put the brakes on," the activation of the pathways.

Genetics of Tumor Suppressor Genes

Mutations in tumor suppressor genes can arise spontaneously by exposure to a mutagenic substance such as ultraviolet light or certain chemi-

mitosis separation of replicated chromosomes

tumors masses of undifferentiated cells; may become cancerous

cell cycle sequence of growth, replication, and division that produces new cells

cals. In such cases, only the mutated cell and its descendants will be affected. Mutations can also be inherited from a parent or arise early in development. In these cases, almost all the cells of the body will inherit the same mutation.

A mutation in a single tumor suppressor gene is usually not enough to cause cancer. This is because each cell contains two copies of each gene, one inherited from each parent. Most cancer-causing mutations cause a loss of function in the mutated gene. Often, having even one functional copy is enough to prevent disease, and two mutations are needed for cancer to develop. This is known as the “two-hit” model of carcinogenesis.

This model was first described in retinoblastoma, a common cancer of the **retina**. The affected gene (called the retinoblastoma gene) is a tumor suppressor. Spontaneous (noninherited) mutations are rare, but since there are many millions of cells in the retina, several of them will develop the gene mutation over the course of a lifetime. It would be very unlikely, though, for a single cell to develop two spontaneous mutations (at least in the absence of prolonged exposure to carcinogens), and thus noninherited retinoblastoma is very rare. When it occurs, it almost always affects only one eye—the eye in which the unlucky doubly hit cell resides.

retina light-sensitive layer at the rear of the eye

If, however, a person inherits one copy of an already mutated gene from one parent, every cell in the eye starts life with one “hit.” The chances are very high that several cells will suffer another hit sometime during their life. The chances are thus very high that the person will develop retinoblastoma, almost always in both eyes, since the necessary second hit is common enough that cells in both eyes will be affected. Because inheriting a single copy of the mutated gene is so likely to lead to the disease, the gene is said to show a dominant inheritance pattern.

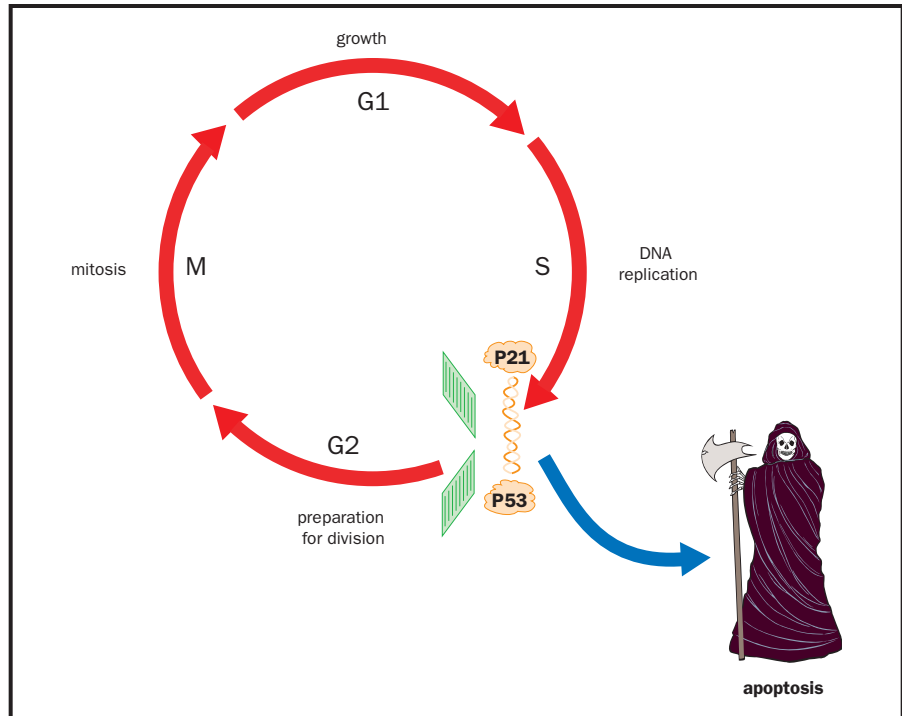
Generalized Tumor Suppressor Genes

There are a growing number of genes that have been identified as having some function as tumor suppressor genes. The table below lists genes and their associated tumor types. One of the most important tumor suppressor

Table 1.

Gene Symbol	Gene Name	Main Tumor Type	Secondary Tumor Type	Chromosomal Location
APC	Adenomatous polyposis coli	Familial adenomatous polyposis of the colon	—	5q21-q22
BRCA1 and 2	Familial breast/ovarian cancer 1 and 2	Hereditary breast cancer	—	13q12.3
CDKN1C	Cyclin-dependent kinase inhibitor 1C (p57) gene	Beckwith-Wiedemann syndrome	Wilms' tumor and rhabdomyosarcoma	11p15.5
MEN1	Multiple endocrine	Multiple endocrine neoplasia	Parathyroid/pituitary	11q13
NF1	Neurofibromatosis type 1 gene	Neurofibromatosis type 1 syndrome	Neurofibromas, gliomas, pheochromocytomas and myeloid leukemia	17q11.2
NF2	Neurofibromatosis type 2 gene	Neurofibromatosis type 2 syndrome	Bilateral acoustic neuromas, meningiomas and ependymomas	22q12.2
TSC1	Tuberous sclerosis type 1	Tuberous sclerosis	Some hamartomas and renal cell carcinoma	9q34
TSC2	Tuberous sclerosis type 2	Tuberous sclerosis	Some hamartomas and renal cell carcinoma	16p13.3

Tumor suppressor genes act as gatekeepers for passage from S (synthesis) phase to G2 and mitosis. Activation of these genes, which may occur if DNA replication cannot be successfully completed, triggers apoptosis, or programmed cell death.



genes is *TP53* (more commonly known as *p53*). This gene was originally identified as a germ-line mutation in the rare inherited cancer called Li-Fraumeni Syndrome, but it has since been shown to be involved in a wide variety of cancer types. The *p53* gene is lost (e.g., the gene is deleted from the chromosome) in about 50 percent of all cancerous cells.

The *p53* protein is responsible for controlling the cell cycle checkpoint at the stage where the cell makes a decision to duplicate its genome, called the G2/S boundary. Along with p21 (another essential protein at this boundary), *p53* protein monitors the state of the DNA to ensure that the genome is intact and not damaged. The S phase is where the genome is duplicated to get ready for cell division, so it is important that any damage and errors be repaired. If the cell is unable to repair the damage to its DNA, *p53* can induce the programmed cell death pathway (called apoptosis) that kills off the cell, thus preventing division of a cell with damaged DNA. If *p53* is not functional, the cell cycle is not arrested and any errors will be duplicated and passed on when the cell divides.

Mechanisms of Functional Tumor Suppressor Loss

There are three main ways in which a cell can lose the functionality of its tumor suppressor genes. Chromosomal aberrations, such as balanced reciprocal translocations, can occur. In such translocations, two unlike chromosomes switch segments. The most common such aberration is the chromosome 11 and 22 t(11;22) (q23;q11) translocation. It occurs in 10 to 15 of every 10,000 newborns and is the most common cause of childhood leukemia. The chromosome 9 and 22 t(9;22)(q34;q11) translocation gives rise to the characteristic derivative of chromosome 22, called the Philadelphia chromosome after the city where it was first found, and results in chronic myelogenous leukemia.

Constitutional chromosomal aberrations, which include deletions and **aneuploidy**, are sometimes associated with an increase in specific kinds of cancers. For example, a deletion on chromosome 13 band q14.1 is associated with retinoblastoma. Trisomy 21 in individuals with Down syndrome is associated with a 1 percent occurrence of leukemia.

aneuploidy abnormal chromosome numbers

Viral oncoproteins can interact with tumor suppressor gene proteins. The human papillomavirus (HPV) is a small DNA virus that causes warts. Various subtypes of HPV are associated with cervical cancer. The viral transforming protein E7 has the ability to interact with the retinoblastoma protein, thus interfering with the cell cycle checkpoint controlled by the retinoblastoma protein. Similarly, another HPV gene, *E6*, interacts with the *p53* gene, causing the degradation of the p53 protein, thus allowing the cell cycle to go unchecked. SEE ALSO APOPTOSIS; BREAST CANCER; CANCER; CELL CYCLE; CHROMOSOMAL ABERRATIONS; COLON CANCER; ONCOGENES.

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Twins

Twins are siblings carried together in the womb and born at the same time. Similarities and differences between twins can be used to answer questions about the role genes and the environment play in the development of traits such as personality, intelligence, and susceptibility to disease. While results from any single pair of twins cannot provide conclusive answers to such questions, the study of large numbers of twin pairs allows researchers to draw conclusions about inheritance with a significant degree of confidence.

Developmental Mechanisms

Twins are classified as either dizygotic or monozygotic. Dizygotic twins (also called fraternal twins) arise from two separately fertilized eggs, or zygotes. In humans, usually only one egg is released at a time from a woman's ovaries. When two are released, both may become fertilized by separate sperm and implant in the uterus. Dizygotic twins develop separate placentas and amniotic sacs. They may be of the same or different sexes. In the absence of reproductive technology interventions, dizygotic twinning occurs in approximately three of every thousand human births, a rate that increases with maternal age, varies with ethnic group, and is probably influenced by genes that control pituitary function. Various types of assisted reproductive technologies routinely create dizygotic twins, triplets, and higher numbers of offspring.

Monozygotic twins (also called identical twins) arise from a single fertilized egg. At some point after the zygote begins to divide, the cell mass splits into two, creating two embryos from one. Monozygotic twinning



Monozygotic, or identical, twins occur in one-quarter of one percent of all human births and begin life with the same set of genes.



occurs in approximately 0.25 percent of human births. Monozygotic twins are always of the same sex. If the cell mass splits before about day five after fertilization, the two embryos will develop with separate placentas and separate amniotic sacs. This occurs in about two-thirds of human monozygotic twins. Between day five and about day nine, splitting leads to two amniotic sacs but one placenta. This occurs in about one-third of Monozygotic twins. Twins that split after day nine will share the amniotic sac. Splitting that late also increases the likelihood that the twins will not separate completely and will develop into conjoined (Siamese) twins.

Monozygotic versus Dizygotic Twins

Because monozygotic (MZ) twins develop from a single fertilized egg, they begin life with exactly the same set of genes. In this respect, they are clones—organisms whose genes are identical. As discussed below, however, they may accumulate genetic and other differences during development.

In contrast, dizygotic (DZ) twins are no more genetically close than any pair of siblings. While it is commonly said that siblings share half their

genes, this is incorrect for two reasons. First, the random nature of meiosis and fertilization means that two siblings could end up with many, or few, genes from a particular parent in common. Second, there are many human genes for which there is only one common form, or allele. Therefore, any two people will share many alleles, regardless of their relationship. Only those genes with more than one allele form the basis of human genetic variation. These are the real focus of the question about gene-sharing in siblings. Of these variable genes, siblings (including dizygotic twins) on average share half.

Because dizygotic twins are the same age, they may share more of their environment than would two siblings of different ages. For instance, because they are likely to be engaged in similar activities, dizygotic twins are more likely to have similar environmental exposures (including behaviors, diet, hobbies, exposure to infectious agents, and exposure to chemicals)—whether at home, at school, or in the community—than two siblings of different ages and different activity patterns. It is this similarity of environment but difference of genes that makes them a useful contrast to monozygotic twins, whose environments and genes are largely identical.

Similarities and Differences between Monozygotic Twins

The fertilized egg cell that gives rise to MZ twins begins life with a single set of genes, and so we might predict that every cell that arises from it would be exactly identical. However, small differences between daughter cells may accumulate throughout embryonic development and later in life. The earliest difference may be in the **mitochondria** each inherits. Mitochondria are the cell's power plants and contain a small amount of DNA. Some of the hundreds of mitochondria in a cell may contain mutations. If the cells that create the two twins carry different mitochondrial genes, even identical twins will be genetically different. Mutations can also accumulate during embryonic development, or after birth, either in the mitochondrial genes or the genes in the nucleus. Such mutations may have a significant effect: Some types of cancer are due to mutations accumulated during one's lifetime, often through exposure to environmental chemicals or radiation.

mitochondria energy-producing cell organelle

For the vast majority of genes, though, MZ twins are exactly identical. Nonetheless, twins do experience slightly different environments, even when reared together, and any early differences between them may be accentuated by families members, or by one another, leading to development of very different personalities.

Amazing Twin Similarities

Some of the most tantalizing clues to the genetic basis of human personality and behavior come from studies of MZ twins reared apart since birth. Such twins have the same genes but, presumably, different environments. A major study of more than 100 such twin pairs showed some remarkable coincidences. A pair of twins meeting for the first time at age thirty-nine each arrived wearing seven rings, two bracelets on one wrist, and a watch and one bracelet on the other wrist. Another twin pair discovered they each had dogs named Toy, had married and divorced women named Linda, remarried women named Betty, and named their sons James Allan and James Alan.



Table 1.

CONCORDANCE IN TWIN STUDIES**Pairwise concordance**

Number of twin pairs in which both are affected
Total number of twin pairs

Proband-wise Concordance

$$\frac{[2c_2 + c_1]}{[2c_2 + c_1 + d]}$$

A proband is an independently ascertained twin with the disease; independently ascertained means the twin was NOT identified through the co-twin.

c_2 = the number of concordant pairs in which both twins are probands

c_1 = the number of concordant pairs in which only one twin is a proband

d = the number of discordant pairs

Using concordance patterns to estimate the relative contributions of genetic and environmental determinants to a condition or disorder:**If MZ concordance = 100%**

Only genetic determinants likely

If MZ > DZ concordance

Genetic determinants important

Environmental modifiers likely

If MZ concordance = DZ concordance

Shared environmental determinants likely

While these coincidences are amazing, it is important to remember that many more twin pairs in this study did not have such parallel lives or habits. Such stories are curious and provocative but cannot by themselves tell us about the relative contributions of genetics and the environment in shaping personality, behavior, health, or other aspects of the self.

Twin Studies and Concordance

Insight into such questions can be gleaned by several types of studies that compare twins. Comparison of MZ twins reared apart is one type of study but is hampered by the extreme rarity of such twin pairs. Another type of study, comparing MZ twins to DZ twins, is more commonly done, because there are many hundreds of thousands of such twin pairs worldwide. Data on twins have been collected by numerous research groups who have created large and growing databases (registries) that can be mined for information.

Determining a characteristic called concordance plays a crucial role in most such studies. A twin pair is said to be concordant for a trait if both members show it. If neither twin shows the trait, the pair is also concordant, but for the absence of the trait. For instance, twins are concordant for Alzheimer's disease if both develop it. They are discordant if one does have the disease but the other does not.

If a trait is strongly influenced by genes, more MZ twin pairs should be concordant than DZ twin pairs, because MZ twins share more genes. Comparing concordance rates between the two groups, and applying some mathematical analysis, allows researchers to estimate the genetic contribution to a trait, as shown in Table 1.

PAIRWISE CONCORDANCE FOR PARKINSON'S DISEASE

	Concordant Pairs		Discordant Pairs		Pairwise Concordance		Risk of Concordance if MZ
	MZ	DZ	MZ	DZ	MZ	DZ	
Overall	11	10	60	80	15.5%	11.1%	1.39 (0.63-3.10)
First twin diagnosed ≤ 50	4	2	0	10	100.0%	16.7%	6.00 (1.69-21.3)
First twin diagnosed > 50	7	8	58	68	10.8%	10.5%	1.02 (0.39-2.67)

Table 2. The data in the last column indicate the relative risk (RR) and 95 percent confidence interval (CI) for concordance of Parkinson's disease in MZ twins. The risk to the second twin is much higher if the first is diagnosed before age fifty, indicating a strong genetic component in early-onset PD.

Twin Registries

Twin studies can have several starting points. Some investigators begin simply by trying to identify twins who will volunteer to be part of a particular research study. Often twins are sought by advertising for twins with the particular disease of interest. This approach has the advantage of simplicity, as twins identify themselves to the research team.

However, twins who volunteer may differ in some important way from those who do not volunteer, and this could affect the conclusions drawn from the study. For example, MZ twins are more likely to volunteer, in general, than DZ twins are. This tendency to volunteer for twin studies among MZ twins is probably because being a twin is a more central part of the identity of MZ pairs than DZ pairs. Also, twins concordant for a particular disease are more likely to volunteer than those without the disease are. If both influences are at work in the same study, more concordant MZ twins than DZ twins may be identified, not because there is an actual difference in concordance between MZ and DZ twins (and thus a genetic effect at work), but because more concordant MZ twins volunteered for the study. If this pattern of volunteerism is mistaken to represent the true pattern of the disease in all twins, an inappropriate conclusion that the disease has genetic causes could result.

Other twin registries attempt to identify all twins within a particular population. One approach is the statewide or national twin registry. All twin births in the region are reported to a central registrar. This results in a more complete picture of all twin pairs in these populations. Examples include the statewide Virginia and Minnesota twin registries in the United States and many national twin registries, including those in the United Kingdom, Australia, the Scandinavian countries, Germany, Belgium, the Netherlands, Italy, and Sri Lanka.

Twin registries have also been assembled from among special populations. Examples in the United States are registries assembled from military records (the World War II Veteran Twins Registry and the Vietnam Era Twin Registry) and from Medicare files (the U.S. Registry of Elderly African-American Twins). In these registries, likely adult twins were identified by searching records to identify individuals with identical dates of birth, birthplaces, and surnames. These individuals were then contacted to

verify whether they actually constituted a twin pair. Registries may also be established by identifying twin births within a health maintenance organization (such as the Kaiser Permanente Twin Cohort, in California).

Each registry varies in the amount of contact with registrants. In all, individual contact is strictly monitored to preserve the privacy of each twin. Every research proposal must be approved by a panel to assure the scientific value of the project, the justification for doing the study in twins, and to ensure that the privacy and safety of individual twins will be protected.

Twin registries can be useful starting points for investigating many questions about the genetic and environmental determinants of a trait. Records linkage studies involve no personal contact with the twins. Instead, information in the twin registry is “linked” electronically to information in another database, such as a national health insurance database or a cancer registry. In this way, twins with a particular health problem can be identified, and concordance estimates can be calculated. Similarly, information collected for each twin at registration can later be used to investigate certain kinds of questions without ever contacting the individual twins. On the other end of the spectrum, twins may be asked to volunteer for physical examinations, blood tests, radiological studies, or interviews. Depending on the questions asked, such studies may be useful for comparing concordance, or for identifying risk factors or modifying factors for a trait.

WHY STUDY TWINS?

- To estimate the relative contributions of genes and environment to the cause of disease by comparing MZ to DZ concordance;
- To investigate environmental determinants of etiology in discordant twin pair studies;
- To investigate environmental influences on disease course in concordant twin pair studies;
- To characterize “presymptomatic” or “at risk” states by studying the unaffected twins in discordant pairs.

Twin Studies to Investigate the Cause of Parkinson’s Disease

An example of the use of investigations in twins to understand more about a disease is provided by recent work in Parkinson’s disease. Parkinson’s disease (PD) is a progressive neurodegenerative disease causing slowness, tremor, and problems with walking and balance. PD is rare before age fifty but becomes more common thereafter, with increasing age. The cause of PD has long been debated. Both genetic and environmental causes have been suggested, but neither has been definitively shown. Researchers turned to studies in twins to determine the relative contribution of genes and environment to the disease.

The first studies identified twin pairs by recruiting through physicians and PD patient organizations. Studies in the United States, the United Kingdom, and Germany identified 103 pairs, of which only thirteen were concordant for PD. In Finland, forty-two twins with PD were identified by records linkage, but among these was only one concordant pair—a DZ pair. No study had convincingly demonstrated greater monozygotic than dizygotic concordance for the disease, and in all studies the preponderance of twin pairs were discordant for disease. These findings supported an environmental cause of PD. Nonetheless, the advent of molecular genetics prompted great interest in investigations of genetic causes of disease and prompted the resurgence of the hypothesis that all PD had a genetic cause. To address this, a study in a large, unselected cohort—the National Academy of Sciences/National Resource Council (NAS/NRC) World War II Veteran Twins Registry—was undertaken.

In the mid-1950s, the Medical Follow-up Agency of the Institute of Medicine of the NAS/NRC established a registry of approximately 32,000

Caucasian male twins, all of whom were born between 1917 and 1927 and were veterans of the U.S. Armed Services. In all, 161 twin pairs were identified, twenty-one of which were concordant for PD, as shown in Table 2. In those few pairs with early-onset PD, concordance was greater in MZ pairs. In those with more typical PD, beginning after age fifty, there was no difference in MZ and DZ concordance.

These findings suggest a strong genetic determinant for early-onset disease but predominantly environmental causes in more typical late-onset disease. One caveat is the narrow age range of the twins, who were sixty-seven to seventy-seven years old when studied. Since PD is a late-life disorder, PD in some twins may have been missed with an examination at only one time point. To overcome this, a second evaluation is in progress.

Risk-Factor Investigations in Twins

Studies of twin pairs discordant for disease can be useful for identifying risk factors for disease. Since both genetic and environmental factors are extensively shared by twins, particularly by MZ twins, case-control studies can be particularly powerful. In such a study, each twin is interviewed with regard to specific environmental factors—such as occupation, lifestyle factors, illnesses or injuries, and diet—prior to the onset of the disease in the affected twin. The presence of these factors in the twin with the disease is compared to the twin without disease. An association of an environmental factor with the disease suggests this factor may be causally related. Factors more common among the unaffected twins suggest that the factor may protect against the development of the disease.

Environmental influences on PD have been investigated by studying discordant twin pairs. PD has repeatedly been found to be more common in people who do not smoke cigarettes. Some have proposed that some people are genetically predisposed to both Parkinson's disease and smoking, while others suggest cigarette smoking somehow prevents the degeneration that leads to PD. In two studies of discordant twin pairs, cigarette smoking was more common in the twin without Parkinson's disease, especially in the MZ pairs. Because monozygotic twins are genetically identical, this pattern tips the scales in favor of a direct biological action of cigarette smoke.

As medicine focuses more on early intervention or prevention, it becomes important to identify those persons at risk for a particular condition. This can be a problem if there is no diagnostic test. In discordant twin pairs, the unaffected twin is more likely to be “at risk” for a particular condition, whether due to shared genes or environment, than would be true for two nontwins. Therefore, studying the unaffected “at risk” twin may help to clarify what features are useful for predicting those who later will develop a particular disease. For example, in the PD twin study, the unaffected twins are being studied prospectively with brain imaging tests that may show early evidence of injury to the brain area damaged in PD. If abnormalities on this test are found to precede the development of PD, this could provide a useful method of early detection. When treatments to slow or stop the onset of PD are available, individuals with imaging abnormalities may receive intervention before symptoms develop.

